

The rationality and structure of AI investment - a case study of OpenAI

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Abstract

Over the last few years, the investment in AI companies has grown rapidly, but they show escalating losses. They are not profitable yet and continue to operate at a net loss. This paper examines whether investment in such companies is rational, using OpenAI as a case study. The core method used in the study is to compare the company's internal projections with its realized outcomes through 2026. The study draws on two types of sources, primary and secondary. These include OpenAI's internal projections reported by such journals as *The Wall Street Journal*, *The Financial Times*, *The Information*, the company's own public announcements, and peer-reviewed papers. The case analysis found that its predicted profitability targets slipped from 2029 to beyond 2030, spending grew faster than anticipated, and projected new revenue lines largely failed to materialise. Moreover, the findings suggest that OpenAI's valuation is based less on its actual financial results than on the expectation that it will dominate the AI market in the future, so the rationality of AI investment relies on assumptions that the company's own record does not yet support.

Keywords: OpenAI, case study, investment, profitability, financial analysis.

1. Introduction

There has been a wide adoption of AI systems across many industries and companies, including manufacturing, medicine, energy and more, and AI is reshaping how organizations operate, compete, and scale. For example, in biotech and pharmaceutical sectors, AI has significantly enhanced the identification of new drug-target combinations (Cao et al., 2024). In retail, it has been found that the adoption of intelligent shelf monitoring in a fast moving consumer goods sector has boosted its product sales (Cao et al., 2024). According to McKinsey's March 2025 Global Survey, 88 percent of organizations now use AI in at least one business function, up from 78 percent a year earlier (Singla et al., 2025).

Consequently, those companies that provide AI services are making a massive amount of revenue, which, in turn, is attracting vast amounts of investments to those companies. However, these investments have not turned to profit yet. In fact, most companies are spending more money than they are earning.

Existing research claims the companies expect to spend substantially more money in the future while making less revenue. For example, according to internal financial projections reviewed by *The Wall Street Journal*, OpenAI does not expect to turn a profit until 2030 (Jin et al., 2026). However, there is a lack of case studies investigating the financial performances and prospects of specific AI companies. Therefore, this paper aims to examine the reasons behind significant and consistent investments into AI companies, specifically OpenAI. The key questions guiding this study are: Is the continued investment in OpenAI rational, given the company's lack of profitability, sliding break-even dates, and the challenges of the training-cost treadmill? Additionally, how do investors justify these valuations against OpenAI's own projections, and is that justification sound?

The primary goal of this paper is to understand whether AI investment will ever turn into profit or not. This paper contributes to the broader debate on the rationality of AI investments.

2. Literature review

2.1. How AI has been changing societies and industries

AI has achieved or even surpassed human performance in several areas (Co et al., 2024). For example, in strategic games like chess and GO, AI has achieved superhuman performance. In 1997, Deep Blue defeated the world champion and one of the best chess players of all time Garry Kasparov in a chess match. Since then AI performance in chess continued to progress beyond human level in the following two decades, and today, chess players still cannot defeat the world's strongest chess engine, Stockfish.

Suleyman (2023) noted that Go is an ancient East Asian game considered to be even more complex than chess. After just three pairs of moves in chess there are about 121 million possible configurations of the board. But after three moves in GO, there are on the order of 200 quadrillion possible configurations. In March 2016, AlphaGO, an AI program developed by Google, was pitted against Lee Sedol, a virtuoso world champion. It went on to beat Sedol 4-1.

Cao et al. (2024) also emphasized the role of AI in the financial investment area, claiming that investors adhering to AI advisors' recommendations outperform others on peer-to-peer lending platforms. A comparison of an AI model, which was trained using data from Prosper.com, to human investors revealed that the AI system surpasses crowd investors in accurately forecasting loan default probabilities. Robo-Analysts' investment recommendations not only exhibit a more balanced distribution and less bias towards glamour stocks than human analysts, but also yield substantial long-term investment benefits.

Farhan et al. (2024) noted the importance of AI in education in his case study. The results showed a strong correlation (correlation coefficient of 0.719) between the use of AI techniques and the quality of educational services. AI technologies

contribute to enhancing the quality of education by providing personalized learning for each student, effectively analyzing data to improve teaching strategies, and saving time and effort for teachers. They are also used in predicting the performance of students, automating evaluations to enhance objectivity, analyzing students' learning processes using educational robots, and identifying factors that enhance the appeal of classes. In terms of cost, despite the comparatively high cost of implementing AI in colleges, it is considered cost-effective compared to the expenses associated with manual labor. Utilizing AI in the long run among university students is more cost-effective compared to traditional educational approaches that involve manual tasks.

Over the last few years, AI's role in companies' profit transformation has become indispensable. It is helping them to make more profit by making better decisions, minimising expenses, and improving efficiency within the company. According to Harb (2024), a 2023 McKinsey Global Survey found that companies with complete AI integration had profit margins of 20%-30%. Moreover, AI-adopting companies reduced ROI time frames by 80%, and 89% of firms achieved ROI within 18 months of adoption.

2.2. Recent AI trends and investment trends

Thus, it is not wrong to assume that companies providing AI services are attracting massive amounts of investments and making enormous amounts of revenue. For example, OpenAI has attracted roughly \$201 billion of investments in 10 years, Anthropic \$128.7 billion in 5 years, and xAI \$42.9 billion in 3 years (according to crunchbase.com). While there are a lot more companies that have attracted AI investments, these are some of the most popular and most-invested companies.

Interest in investments in AI companies has especially grown over the last few years. In 2020, \$75 billion was invested in AI ventures by venture capital investments. This accounted for only 20% of all VC investments that year (Montanaro et al., 2024). In 2025, venture capital investment in AI ventures made up around \$258.7 billion, which accounts for 61% of all VC investment (\$ 427.1 billion). OpenAI attracted \$122 billion investment in the first half of 2026 alone, which accounts for more than 50% of all investment it received since its founding. The AI market reached \$757.58 billion in 2025 and is projected to hit \$1,807.84 billion by 2030, a compound annual growth rate of roughly 19%. When looking at the end-user, direct subscriptions, and tool licensing, the AI market stood at \$244 billion in 2025 and is projected to hit \$1,200 billion (Eze et al., 2025).

2.3. Financial prospects of these AI investments

One of the reasons for this dramatic increase in AI companies over the past years was its distinctive GPT (general purpose technology) characteristics

(Montanaro et al., 2024). The AI domain may be an attractive area for VC investment because of the significant technological benefits and learning opportunities that may be generated by AI ventures to the advantage of customers and/or incumbents in the market (Montanaro et al., 2024).

Montanaro et al. (2024) presented three reasons for why VC investment to AI isn't as high as investment in the non-AI 4.0 domain (digital systems that do not use AI): first, AI ventures are generally subject to stronger information asymmetries compared to ventures operating in other digital areas because AI is an emerging, complex, and rapidly evolving technology; second, AI brings with it several regulatory and competitive challenges that should not be overlooked, and that do not impact the non-AI 4.0 industry as much. The rapid pace of AI developments is accompanied by severe competition and regulatory risks that could hinder AI ventures in achieving long-term success and VCs in reaping the benefits of their investments; third, the unpredictability of unchecked technological developments that may lead to ethical and privacy concerns and the vulnerability of AI-systems to cyber-attacks could cause potential losses to investors. In the years since, however, AI technology has dominated such a great part of the economy and society that it is becoming challenging to predict the rate of investments with greater precision.

3. Case study of OpenAI

3.1. Background information

OpenAI was selected as the case study for three reasons. It has attracted the highest amount of investment among any other companies, including those in the AI sector. Secondly, it has shown higher losses, revenue, and spendings than any other AI company. Finally, OpenAI is unusual among AI companies in the fact that there is a lot more financial data available about it than any other AI company. These reasons make OpenAI an interesting and reasonable case to do research on.

In 2024, Open AI said its model will improve and its gross margin will hit around 49% in 2025 (Weinberg, 2024). OpenAI projected steady improvements in the coming years and stated to hit around 67% gross margin by 2028 (Weinberg, 2024). However, the gross margin fell from 40% to 36% in 2025, missing its forecast of 49% (Muppidi, 2026). One of the main culprits was a jump in inference costs, the payments they make to cloud providers to power AI models when customers talk to their chatbots or use models through an application programming interface (Muppidi, 2026). This highlights that the costs of running AI models are climbing and that OpenAI's aim of hitting 67% gross margin by 2028 may not be achieved. Even if the gross margin did hit 49% in 2025, it would still be far below the typical gross margin of an AI startup that usually has 65-70% gross margin (Weinberg, 2024).

Sora was OpenAI's text-to-video model, which could generate a video while adhering to the user's prompt. It was shut down on April 26 after less than 2 years of its release. The main reason for its closure was the cost of running it. Jin and

Toonkel (2026) stated that Sora was losing roughly \$1 million a day. Unlike language models, which learn from text, video models have to make sense of entire moving worlds, making them far more expensive to develop and run. AI chips are the most precious commodity at any leading research lab, and at OpenAI, Sora was eating up far too many of them. “It was a resource black hole with limited monetisation” told the Forrester analyst Thomas Husson (Chia & Calder, 2026). Plus, the model didn’t improve the capabilities of its language models. As Jin and Toonkel noted, another reason for Sora’s shut down was that the user interest in the model has faded over the months. When the stand-alone Sora app was launched in September 2025, Altman likened it to the moment when the company first released ChatGPT since the worldwide user count peaked at around 1 million. But it never reached that level again. Sora was more AI slop than AI magic. Usage flatlined at 500,000 by the end of 2025.

Khan (2024) stated that OpenAI unveiled a new team dedicated to ensuring that future AI systems that might be more intelligent than all humans combined could be safely controlled. To signal how serious the company was about this goal, it publicly promised to dedicate 20% of its then-available computing resources to the effort. Less than a year later, that team was disbanded amid staff resignations and accusations that OpenAI is prioritizing product launches over AI safety. The company never fulfilled its commitment to provide 20% of its computing power. Instead, the team repeatedly saw its requests for access to graphics processing units, the specialized computer chips needed to train and run AI applications, turned down by OpenAI’s leadership. After resigning, Leike, who co-lead the team, published a series of posts on X in which he criticized his former employer, saying “safety culture and processes have taken a backseat to shiny products”. This not only raises questions about how serious the company is about safety issues, but also about how committed OpenAI ever was to honoring its public pledge, and whether other public commitments the company makes should be trusted.

According to OpenAI’s own projections in 2024, the company won’t turn a profit until 2029, when its revenue would hit \$100 billion (Weinberg, 2024). However, according to OpenAI’s own projections in 2026, the company doesn’t expect to make profit even until the 2030s (Jin & Rattner, 2026).

3.2. Financial Analysis

Since its founding, OpenAI has attracted around \$201 billion of investment, while making no profit. So far, it has only been spending its money and investment. In 2025, OpenAI spent around \$34 billion, while earning \$13 billion (Hammond & Elder, 2026). Of those \$34 billion, \$19 billion was spent on research and development of new models (Hammond & Elder, 2026). Without the development of new models, OpenAI’s spending is not that large. In fact, according to their estimations, they are already making a small profit in 2026 without considering R&D cost. However, it can’t stop developing new models, as competitors ship better models every year. So, the company will have to pour more and more money into

new-model development, which is becoming considerably more expensive to do year by year.

With this idea, where OpenAI will keep pouring money into developing new models, OpenAI will turn into profit in 2030, according to its own projections in 2026. But, in 2024, OpenAI projected that it will turn into profit in 2029. Its timeline for profitability may even change further to the 2030s because its projections are not going as planned. In 2024, it projected its gross margin to increase from 40% to 49% in 2025, but instead of increasing it fell to 36%.

Moreover, Sora had turned into a liability for OpenAI in the months after its release(Jin & Toonkel, 2026). It made \$1.4 million in global net in-app revenues, while losing \$1 million a day. Because of Sora, OpenAI has so far lagged behind its startup rival Anthropic in winning the AI market, imperiling its lead in the AI race (Jin & Toonkel, 2026). If the company develops new models that will fail like Sora, making it lose money, its projection to turn a profit in 2030 may change further.

Table 1. Open AI metrics

Metrics	2022	2023	2024	2025	2026*
investment	\$2 billion	\$10.3 billion	\$20.1 billion	\$47.6 billion	\$122.075 billion
R&D	N/A	N/A	\$7.81 billion	\$19 billion	N/A
new products	Whisper	GPT-4V(ision) system card,	GPT-4o mini	GPT-5.3-co dex	GPT 5.5
losses	\$540 million	\$1.5 billion	\$5 billion	\$39 billion	N/A
Revenue	\$28 million	\$1.6 billion	\$3.7 billion	\$13.07 billion	N/A

* Assessed as of July 2026

Table 2. Anthropic metrics

Metrics	2022	2023	2024	2025	2026*
Investment	\$580 million	\$4.15 billion	\$6 billion	\$20 billion	\$95 billion
R&D	N/A	\$400 million	N/A	\$4.1 billion	N/A
New products	N/A	Claude 2	Claude 3.5 Sonnet	Sonnet 4.5 and Claude Code	Sonnet 5 and Fable 5

Losses	\$500 million	\$1.5 billion	\$5.6 billion	\$5.2 billion	N/A
Revenue	\$10 million	\$100 million	\$1 billion	\$6 billion	N/A

** Assessed as of July 2026*

As can be seen from Tables 1 and 2, both OpenAI and its closest competitor Anthropic show around the same figures. Both companies have consistently been showing substantial losses and relatively small revenues, but it didn't stop them from receiving extensive investment. As the years progressed and even though losses increased, the companies still got massive amounts of investment. In the first half of 2026, OpenAI received around \$122.075 billion and anthropic around \$95 billion of investment, and as can be seen, a massive amount of that spending was on R&D cost. As for the losses, they have been showing similar amounts since 2022, but OpenAI's losses soared from \$5 billion in 2024 to an immense \$39 billion. The majority of the losses, which was around \$30 billion, could be explained by a non-cash accounting charge linked to the company's previous structure rather than its underlying operations. However, even if that loss wasn't calculated, the company would still have around \$9 billion in loss, which was substantially higher than its loss in 2025.

4. Conclusions

This paper explored the rationality of investment into OpenAI. In sum, OpenAI has failed to reach its own projections many times for different reasons, but mainly because of an increase in the cost of hardware for data centers. It is possible that, in the future, the cost may increase further and disrupt the company's plans and projections. The company may also lose money if their models become unsuccessful, as was the case with Sora, which may also disrupt its projections. It is also clear that the company needs all of its computing power for model development and doesn't want to allocate it for safety research, which is something that everyone should be concerned about. Finally, the company's projection of becoming profitable in 2030 is highly likely to be changed, as it has repeatedly failed to meet its earlier projections and its publicly stated promises. It is likely the company will burn through a giant amount of cash in the coming years and is counting on its investors to help keep the business afloat.

The study has two limitations. First, as a single-case study, its findings describe OpenAI and may not be generalized to the wider AI sector. Second, the subject is fast-moving: all data reflect information available as of July 2026, and subsequent disclosures may alter the picture.

Future research could extend this analysis to other AI companies, such as Anthropic and xAI, comparing their financial projections with realized outcomes.

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